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Extension of the $A^1\Sigma^+ - X^1\Sigma^+$ System of ^{7}LiH Near the Dissociation Limit

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The absorption spectra of ^{7}LiH have been photographed in the 3,500–2,900 Å region using the second order of a 3.4 M Ebert spectrograph. Observations comprise previously observed $A^1\Sigma^+ - X^1\Sigma^+$ transitions and several new vibrational bands of this system near the ionisation limit. Rotational and vibrational analysis of these bands ($16 \leqq V' \leqq 20$) have been carried out and the rotational constants for the upper states have been determined.

Introduction

Considerable interest has been shown since 1930's both experimentally and theoretically in the studies of the LiH molecule, being the simplest diatomic metallic hydride [5]. The ground state electronic configuration of LiH is $K(2s\sigma)^2\,^1\Sigma^+$ and the first excited state arises by promoting an electron to the next unfilled $(2p\sigma)$ molecular orbital. In this excitation the principal quantum number remains unchanged therefore the resulting states are designated as non-Rydberg or valence states. The excitation scheme is:

$$K(2s\sigma)^2\,X^1\Sigma^+ \rightarrow K(2s\sigma)(2p\sigma)\,A^1\Sigma^+.$$

The pioneering work on the spectra of LiH corresponding to the $A^1\Sigma^+ - X^1\Sigma^+$ excitation was performed by Nakamura [7]. Subsequently Crawford and Jorgensen [2] photographed the emission and absorption spectra of LiH and LiD in the 3,200–4,300 Å region at high resolution and reported detailed analysis of this system. Li and Stwalley [6] reinvestigated this system using isotopic Lithium hydride and deutrides and reanalysed the data reported by Crawford and Jorgensen [2]. The Rydberg-Klein-Rees potential energy curves of the $X^1\Sigma^+$ and $A^1\Sigma^+$ states were also reported. Orth and Stwalley [6] extended the observed spectral region to 8,700 Å and found 32 new bands of this system. The observed vibrational structure for the ground state was increased from $v''=5$ to $v''=15$ and new Rydberg-Klein Rees turning points were calculated up to $v''=12$ of the ground state. More recently Ennen et al. [4] observed the vibrational levels up to $v''=26$ of the $X^1\Sigma^+$-state using laser excited fluorescence measurements and reported improved Dunham coefficients of the $X^1\Sigma$ and $A^1\Sigma$ states of LiD. Docken and Hinze's [3] ab initio multiconfiguration self consistent field calculations indicated at least 20 bound vibrational levels for the $A^1\Sigma^+$-state while 15 had been observed [6].

The present experiments were performed to observe the predicted higher vibrational levels for the $A^1\Sigma^+$-state near the dissociation limit. In addition to the previously known bands of this system, five new bands have been identified and the observed vibrational structure for the excited $A^1\Sigma^+$ state has been extended from $v'=15$ to $v'=20$. The vibrational and rotational constants for the $A^1\Sigma^+$ state have been determined.

Experimental

Lithium metal and hydrogen were placed in an evacuated resistance fornace of five feet length and heat-

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ed to temperature between 600 and 800 °C with the hydrogen pressure varying from 400–500 torr. Absorption spectra were obtained by using a high pressure Xenon arc (45 W) for the continous background source. The spectra were photographed on Q_2-Ilford plates using the second order of a 3.4 m Ebert spectrograph with a reciprocal dispersion of 2.4 Å/mm. The exposure time ranged between 10–15 min.
Measurements of the plates were made by comparision with iron arc lines. The wavelength uncertainty for sharp unblended lines is estimated to be ± 0.01 Å. The vacuum wavenumbers of the iron lines were obtained from the table of wavenumbers [1].

Structure and Analysis of the Bands

The prominent features in the absorption spectrum of LiH in the 2,900–3,500 Å region are associated with the vibrational sequence of the $A^1\Sigma^+ - X^1\Sigma^+$ system. Five new bands of the $A-X$ transition have been recorded and analyzed ($16 \leqq v' \leqq 20$) near the dissociation limit. The bands show overlapping R- and P-branches and no distinct band heads. Both the branches nearly have the same intensity and are running parallel to each other due to the smaller rotational constants in the excited state. The vibrational analysis of these bands was performed according to the work of Li and Stwalley [6]. The wavenumbers of the P- and R-branches are collected in Table 1. The assignments of J values were straightforward.

Table 2. Rotational constants of the vibrational levels of the $A^1\Sigma^+$ state

v', v''	$\nu_{v',v''}$	B'_v	D'_v
16.0	31,810.39 ± 0.14	2.477 ± 0.016	4.28×10^{-4}
17.0	32,169.70 ± 0.19	2.427 ± 0.052	4.25×10^{-4}
18.0	32,519.29 ± 0.16	2.346 ± 0.031	4.20×10^{-4}
19.0	32,856.54 ± 0.13	2.254 ± 0.023	4.15×10^{-4}
20.0	33,181.02 ± 0.21	2.138 ± 0.017	4.13×10^{-4}

All values are given in cm^{-1}

The interpretation that the lower state of the transition is the ground $X^1\Sigma^+$ state of the molecule is further confirmed by the agreement of the combination differences $\Delta_2 F''(J)$ with the data known from Li and Stwalley [6].
The excited state rotational constants B'_v and D'_v were obtained using the combination differences:

$$\Delta_2 F'(J) = R(J) - P(J)$$

where

$$\Delta_2 F'(J) = 4B'_v(J+1/2) - 8D'_v(J+1/2)^3$$

A plot of $\Delta_2 F'(J)/(J+1/2)$ versus $(J+1/2)^2$ gives the intercept at $4B'_v$ and has the slope $8D_v$.
In addition to this graphical method we have also used the least squares method to evaluate the rotational constants. The constants thus obtained are

Table 1. Wavenumbers of the P and R branches

$A^1\Sigma^+ - X^1\Sigma^+$ band system of ^{7}LiH

J	(16,0) band		J	(17,0) band		J	(18,0) band		J	(19,0) band		J	(20,0) band	
	R branch	P branch		R branch	P branch		R branch	P branch		R branch	P branch		R branch	P branch
0	31,815.54		0	32,174.44		0	32,524.00	...	0					
1	31,810.49		1	169.95		1	518.60	...	1	32,855.22	...			
2	795.75	31,771.18	2	154.98	32,130.01	2	503.44	32,479.65	2	838.96	...			
3	771.18	736.55	3	130.01	096.51	3	477.22	444.66	3	812.46	32,781.23			
4	736.55	692.38	4	094.29	051.59	4	441.43	400.06	4	775.98	735.57			
5	692.38	638.56	5	049.02	31,997.34	5	395.41	344.54	5	729.14	679.79			
6	638.56	574.75	6	31,994.07	932.61	6	339.17	279.65	6	671.58	614.20	6	32,988.50	32,934.06
7	574.75	501.71	7	929.92	858.34	7	273.72	204.73	7	604.06	538.45	7	919.40	856.91
8	501.71	418.94	8	855.05	774.80	8	197.62	119.90	8	526.63	452.31	8	839.91	769.23
9	417.94	326.81	9	771.18	682.15	9	112.70	026.07	9	439.14	356.11	9	750.45	671.98
10	325.81	226.07	10	...	579.41	10	015.87	31,921.96	10	341.64	250.99	10	650.70	564.54
11	223.24	114.65	11	...	467.60	11	31,910.73	807.44	11	234.07	135.86	11	540.50	446.94
12	112.10	30,994.68	12	31,459.80	345.40	12	795.75	685.31	12	116.79	011.08	12	420.31	320.16
13	30,991.43	865.90	13	337.45	215.75	13	671.16	553.38	13	31,988.96	31,876.72	13	289.66	183.12
14	862.03	728.44	14	205.87	075.97	14	536.94	412.13	14	852.81	733.12	14	149.89	036.64
15	722.56	581.36	15	065.32	30,927.91	15	393.88	261.62				15	31,999.93	31,880.10
												16	840.82	715.11

given in Table 2. The band origins were obtained using the following relation: (Herzberg, 1965)

$$R(J-1)+P(J)$$
$$=2\nu_0+2(B'-B'')J^2-2(D'-D'')J^2(J^2+1).$$

The band origins are collected in Table 2 along with the standard deviation of the mean.

Discussion

The observations and analysis of five new rovibrational bands of the ^{7}LiH molecule provides more information about the nature of the excited state. The last band observed in the present experiment is (20,0) which lies only $\approx 1{,}500\,\mathrm{cm}^{-1}$ below the dissociation limit reported by Zemke et al. [9]. It is worth mentioning that the rotational structure of the observed bands is becoming diffuse with increasing the vibrational quantum number, particularly for the (20,0) band. This diffuseness can be attributed to predissociation or interaction with the $B'\pi$-state which lies at 4.25 eV above the ground state. The vibrational analysis is consistent with the previous studies [6]. The spacing of the band origins decreases as 368.44, 359.31, 349.59, 337.25 and $324.48\,\mathrm{cm}^{-1}$ for $v'=15, 16, \ldots, 20$ respectively. In Fig. 1 we show the behaviour of the $\Delta G'_{(v+1/2)}$ in the excited state. From the $\Delta G'_{(v+1/2)}$ values, it is quite obvious that at least four more vibrational bands are likely to be observeable under special experimental conditions which is in accordance with the theoretical predictions by Docken and Honze [3].

The rotational constants B'_v and D'_v for the newly observed bands also show a similar trend as for the lower bands [6], the B-values decrease with increasing vibrational quantum number. Using our B'_v and D'_v given in Table 2 we have recalculated the combination differences for each band and found good agreement within our experimental error.

In conclusion the new observations will be useful to construct a more accurate potential energy curve for the $A^1\Sigma^+$ state. High resolution absorption experiment with a longer path length and an intenser light source will be necessary to observe the higher vibrational transitions i.e., between $v'=20$ and the dissociation limit of the $A'\Sigma^+$-state.

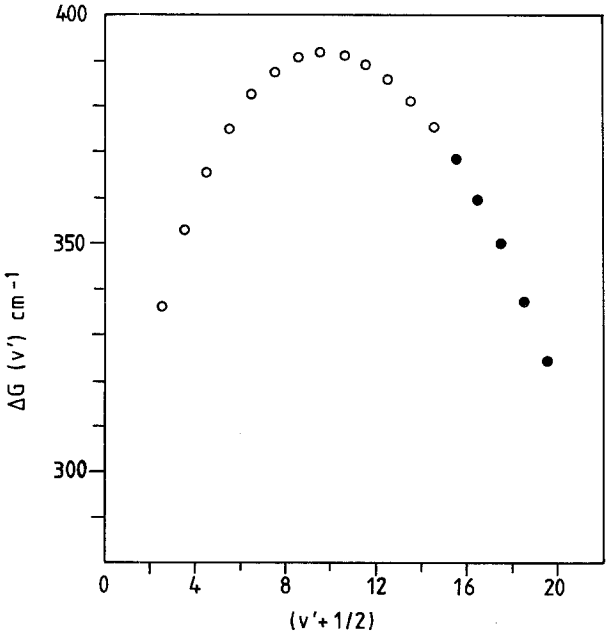


Fig. 1. The variation of the vibrational spacing $\Delta G(v')$ for the $A^1\Sigma^+$ state of ^{7}LiH plotted versus $(v+1/2)$. This figure clearly illustrates the anomalous behavior of $\Delta G(v')$ with increasing vibrational quantum number. The data points up to $v=15$ are taken from Li and Stwalley [6]. The newly observed levels are marked with heavy dots

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